CLASS 8 · PART TEST-1 · STUDENT PERFORMANCE REPORT

# M Jagadeesear Reddy

Pragya G8 (EXI) Achievers · Adm No **246562174**

**Campus** HYD/Nallakunta DD Colony
**JEE Mains** 08 Oct 2026
**JEE Advanced** 09 Oct 2026

## JEE Mains

08 Oct 2026 · 74 questions

**162** / 296 — **54.7%**

| 129 | 89 | 6/63 | 45 |
|---|---|---|---|
| ALL-INDIA | STATE | CAMPUS | PROGRAM |

Weekly-test average **199.2 / 300** (66.4%) → this test **54.7%** ▼ **11.7**

**134 marks lost**: 70 on 14 wrong answers (incl. 14 negative) · 64 on 16 skipped

## JEE Advanced

09 Oct 2026 · 48 questions

**80** / 171 — **46.8%**

| 299 | 204 | 13/63 | 85† |
|---|---|---|---|
| ALL-INDIA | STATE | CAMPUS | PROGRAM |

Weekly-test average **99.8 / 180** (55.5%) → this test **46.8%** ▼ **8.7**

**91 marks lost**: 62 on 15 wrong answers (incl. 10 negative) · 28 on 8 skipped · 1 on partly-correct answers

## 1 Subject scorecard

Your marks in each subject, against the 63 students of this program on campus and against your own weekly-test average.

| SUBJECT | MARKS | SCORE VS CAMPUS | SUBJECT AIR | CAMPUS AVG / BEST | WEEKLY AVG | CHANGE | RIGHT / WRONG / SKIP |
|---|---|---|---|---|---|---|---|
| **JEE Mains** 162 / 296 · 54.7% | | | | | | | |
| Maths | **61**/96 | 64% | **46** | 37.0 / 77 | **73.0**/100 73.0% | ▼ **9.4** | 16 / 3 / 5 |
| Physics | **32**/100 | 32% | **288** | 18.3 / 60 | **72.5**/100 72.5% | ▼ **40.5** | 10 / 8 / 7 |
| Chemistry | **69**/100 | 69% | **275** | 37.4 / 86 | **53.6**/100 53.6% | ▲ **15.4** | 18 / 3 / 4 |
| **JEE Advanced** 80 / 171 · 46.8% | | | | | | | |
| Maths | **36**/54 | 67% | **80** | 24.2 / 41 | **39.0**/60 65.0% | ▲ **1.7** | 10 / 3 / 2 |
| Physics | **12**/60 | 20% | **841** | 9.4 / 40 | **35.5**/60 59.1% | ▼ **39.1** | 5 / 8 / 4 |
| Chemistry | **32**/57 | 56% | **507** | 23.9 / 50 | **25.7**/60 42.9% | ▲ **13.3** | 10 / 4 / 2 |

Bar = your score % | campus average | best on campus  Change = this test % − weekly-test average %

### What this report says

1. **Chemistry** is your strongest JEE Mains subject: 69/100 (69%), All-India subject rank 275.
2. **Physics** needs the most work in JEE Mains: 32/100 (32%), subject rank 288; you lost 15 of its 25 questions.
3. **Maths** is your strongest JEE Advanced subject: 36/54 (67%), All-India subject rank 80.
4. **Physics** needs the most work in JEE Advanced: 12/60 (20%), subject rank 841; you lost 12 of its 17 questions.
5. Biggest fall against your weekly-test average: **JEE Mains Physics** – 32% here against 72% on average.
6. Biggest improvement: **JEE Mains Chemistry** – 69% here against 54% on average.
7. **8 questions** you lost were answered correctly by 70%+ of the batch's top 100 – about **35 marks** within easy reach (Mains Q6, Q28, Q33 · Advanced Q24, Q25, Q36, Q38, Q41).
8. Negative marking cost you **14 in JEE Mains and 10 in JEE Advanced**.
9. You left **16 in JEE Mains and 8 in JEE Advanced** questions blank, and answered 14 in JEE Mains and 15 in JEE Advanced wrong – both are marks to recover.

# JEE Mains – question by question

**162 / 296** · 54.7%
08 Oct 2026

## 2 Every question at a glance

Right · Wrong · Skipped · Deleted (no marks) · Easy miss · Within reach · Tough one

| | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Maths | 1 | 2 | **3** | **4** | **5** | **6** | 7 | 8 | 9 | 10 | 11 | 12 | 13 | **14** | 15 | **16** | **17** | 18 | 19 | ~~20~~ | 21 | 22 | 23 | 24 | **25** |
| Physics | 26 | **27** | **28** | **29** | 30 | 31 | 32 | **33** | 34 | **35** | **36** | 37 | **38** | 39 | **40** | **41** | **42** | **43** | 44 | 45 | 46 | **47** | **48** | **49** | **50** |
| Chemistry | 51 | 52 | 53 | **54** | **55** | 56 | 57 | **58** | **59** | 60 | 61 | 62 | 63 | 64 | **65** | 66 | **67** | 68 | 69 | 70 | 71 | 72 | 73 | 74 | **75** |

## 3 Questions you lost – what to fix first

Every question you got wrong or skipped, one table per subject, easiest marks first. **Top 100** = the batch's top 100 across all campuses; **Campus** = all 63 students of this program on campus.

**Easy miss** 70%+ of the top 100 got it right – easiest marks to win back · **Within reach** 40–69% · **Tough one** below 40%

### Maths

**8** of 24 questions lost · **35** marks · ● 1 easy miss · ● 5 within reach · ● 2 tough one

| Q | PRIORITY | CHAPTER – WHAT IT TESTS | TYPE | YOU | MARKS LOST | TOP 100 RIGHT | CAMPUS RIGHT |
|---|---|---|---|---|---|---|---|
| Q6 | Easy miss | **Quadratic Equations** | Single correct · +4 / −1 | **Skipped** | 4 | **76%** | 38% |
| Q14 | Within reach | **Logarithms & Exponents** – Log-exponential equation; counting solutions | Single correct · +4 / −1 | **Skipped** | 4 | **53%** | 19% |
| Q17 | Within reach | **Surds & Indices** – Cube-root surd identity (matrix match) | Single correct · +4 / −1 | **Skipped** | 4 | **53%** | 41% |
| Q25 | Within reach | **Surds & Indices** | Numerical · +4 / −1 | **Skipped** | 4 | **44%** | 21% |
| Q3 | Within reach | **Sequences & Series** – Product of GP terms; natural-number GP | Single correct · +4 / −1 | **Wrong** | 5 | **42%** | 51% |
| Q4 | Within reach | **Sequences & Series** – Identify AP / GP / HP from a surd expression | Single correct · +4 / −1 | **Skipped** | 4 | **42%** | 19% |
| Q5 | Tough one | **Quadratic Equations** – Range of a rational quadratic expression | Single correct · +4 / −1 | **Wrong** | 5 | **21%** | 6% |
| Q16 | Tough one | **Quadratic Equations** – Roots in GP, condition on discriminant | Single correct · +4 / −1 | **Wrong** | 5 | **19%** | 3% |

## Physics

**15** of 25 questions lost · **68** marks · ● 2 easy miss · ● 5 within reach · ● 8 tough one

| Q | PRIORITY | CHAPTER – WHAT IT TESTS | TYPE | YOU | MARKS LOST | TOP 100 RIGHT | CAMPUS RIGHT |
|---|---|---|---|---|---|---|---|
| Q28 | Easy miss | **Vectors** – Resultant ⊥ smaller vector; find magnitudes | Single correct · +4 / −1 | **Skipped** | 4 | **77%** | 30% |
| Q33 | Easy miss | **Kinematics** – Which v-t graph can show real one-dimensional motion | Single correct · +4 / −1 | **Wrong** | 5 | **75%** | 65% |
| Q29 | Within reach | **Vectors** – Angles between three vectors summing to zero | Single correct · +4 / −1 | **Wrong** | 5 | **62%** | 24% |
| Q40 | Within reach | **Laws of Motion** – Velocity over fixed distance vs mass ($v \propto 1/\sqrt{m}$) | Single correct · +4 / −1 | **Wrong** | 5 | **62%** | 30% |
| Q50 | Within reach | **Laws of Motion** | Numerical · +4 / −1 | **Skipped** | 4 | **44%** | 17% |
| Q27 | Within reach | **Vectors** – Vector components under axis rotation | Single correct · +4 / −1 | **Skipped** | 4 | **43%** | 6% |
| Q35 | Within reach | **Kinematics** – $a = kv^2$ – integrate to get v(s) | Single correct · +4 / −1 | **Skipped** | 4 | **41%** | 5% |
| Q41 | Tough one | **Laws of Motion** – Newton's 3rd law – fan blowing on own sail | Single correct · +4 / −1 | **Wrong** | 5 | **31%** | 11% |
| Q42 | Tough one | **Laws of Motion** – Extra tension when bird leaves the wire | Single correct · +4 / −1 | **Wrong** | 5 | **31%** | 14% |
| Q38 | Tough one | **Kinematics** – Resistance $\propto v^2$ – time inside plank | Single correct · +4 / −1 | **Skipped** | 4 | **27%** | 11% |
| Q36 | Tough one | **Kinematics** – a(t) with modulus – area under a-t | Single correct · +4 / −1 | **Wrong** | 5 | **24%** | 5% |
| Q43 | Tough one | **Laws of Motion** – Normal reaction across an inclined contact | Single correct · +4 / −1 | **Wrong** | 5 | **20%** | 10% |
| Q47 | Tough one | **Vectors** – Net displacement from multi-leg journey | Numerical · +4 / −1 | **Skipped** | 4 | **15%** | 3% |
| Q48 | Tough one | **Kinematics** – Distance from a-t graph | Numerical · +4 / −1 | **Wrong** | 5 | **15%** | 3% |
| Q49 | Tough one | **Kinematics** – Free fall then deceleration – find height | Numerical · +4 / −1 | **Skipped** | 4 | **15%** | 2% |

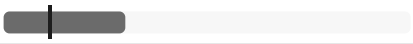
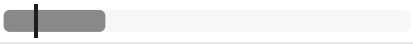
## Chemistry

**7** of 25 questions lost · **31** marks · ● 6 within reach · ● 1 tough one

| Q | PRIORITY | CHAPTER – WHAT IT TESTS | TYPE | YOU | MARKS LOST | TOP 100 RIGHT | CAMPUS RIGHT |
|---|---|---|---|---|---|---|---|
| Q54 | Within reach | **Mole Concept & Basics** – Minimum molecular mass of insulin from 3.4% S | Single correct · +4 / −1 | **Skipped** | 4 | **67%** | 25% |
| Q55 | Within reach | **Mole Concept & Basics** – Laws of chemical combination | Single correct · +4 / −1 | **Wrong** | 5 | **66%** | 17% |
| Q65 | Within reach | **Periodic Classification** – Largest radius among isoelectronic ions | Single correct · +4 / −1 | **Skipped** | 4 | **64%** | 33% |
| Q75 | Within reach | **Chemical Bonding** – Hybrid orbitals + lone pairs in $NH_3$ | Numerical · +4 / −1 | **Skipped** | 4 | **60%** | 30% |
| Q58 | Within reach | **Atomic Structure** | Single correct · +4 / −1 | **Wrong** | 5 | **57%** | 16% |
| Q59 | Within reach | **Chemical Bonding** – VBT, orbital overlap, hybridisation statements | Single correct · +4 / −1 | **Skipped** | 4 | **50%** | 29% |
| Q67 | Tough one | **Chemical Bonding** – Strongest ionic bond (charge and size) | Single correct · +4 / −1 | **Wrong** | 5 | **32%** | 13% |

## 4 Chapter scorecard

Your marks on each chapter's questions against the campus average for the same questions. Weakest chapters first within each subject.

| CHAPTER | QS | YOUR MARKS | YOU VS CAMPUS AVERAGE | CAMPUS AVG | QUESTIONS YOU LOST |
|---|---|---|---|---|---|
| **Maths** | | | | | |
| **Quadratic Equations** | 5 | **6**/20 |  | 11% | **Q5**, **Q6**, **Q16** |
| **Sequences & Series** | 5 | **11**/20 |  | 33% | **Q3**, **Q4** |
| **Logarithms & Exponents** | 3 | **8**/12 |  | 40% | **Q14** |
| **Surds & Indices** | 11 | **36**/44 |  | 53% | **Q17**, **Q25** |
| **Physics** | | | | | |
| **Laws of Motion** | 8 | **8**/32 |  | 7% | **Q40**, **Q41**, **Q42**, **Q43**, **Q50** |
| **Kinematics** | 10 | **13**/40 |  | 21% | **Q33**, **Q35**, **Q36**, **Q38**, **Q48**, **Q49** |
| **Vectors** | 7 | **11**/28 |  | 27% | **Q27**, **Q28**, **Q29**, **Q47** |
| **Chemistry** | | | | | |
| **Chemical Bonding** | 6 | **11**/24 |  | 34% | **Q59**, **Q67**, **Q75** |
| **Atomic Structure** | 5 | **15**/20 |  | 31% | **Q58** |
| **Mole Concept & Basics** | 9 | **27**/36 |  | 32% | **Q54**, **Q55** |
| **Periodic Classification** | 5 | **16**/20 |  | 58% | **Q65** |

Bar = your % of the chapter's marks | campus average

## 5 JEE Mains · Maths – questions to revisit

**8** questions · easy misses first
cropped from the question paper

### Q6 Quadratic Equations

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 4
Top 100: **76%** got it right · Campus: **38%** (24 of 63)

**Easy miss**
**You: Skipped**

6. If the sum of the roots of the quadratic equation $ax^2+bx+c=0$ is equal to the sum of the squares of the reciprocals, then

(A) $ab^2, ca^2, bc^2$ are in A.P.
(B) $a^2b, c^2b, b^2c$ are in A.P.
(C) $ab^2, ca^2, bc^2$ are in G.P.
(D) $a^2b, c^2b, b^2c$ are in G.P.

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

### Q14 Logarithms & Exponents – Log-exponential equation; counting solutions

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 4
Top 100: **53%** got it right · Campus: **19%** (12 of 63)

**Within reach**
**You: Skipped**

14. The equation $x^{(3/4)(\log_2^x)^2+\log_2^x-(5/4)}=\sqrt{2}$ has

(A)at least one real solution
(B)exactly four real solutions
(C)exactly two irrational solution
(D)exactly four non-real solutions

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q17

**Surds & Indices – Cube-root surd identity (matrix match)**

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 4

Top 100: **53%** got it right · Campus: **41%** (26 of 63)

**Within reach**

**You: Skipped**

17. Let $x=\left(a+\sqrt{a^2+b^3}\right)^{\frac{1}{3}}+\left(a-\sqrt{a^2+b^3}\right)^{\frac{1}{3}}$ and $x^3+Abx-B.a=0$, Then match COLUMN-I with respect to COLUMN-II

| | COLUMN-I | | COLUMN-II |
|---|---|---|---|
| I) | A + B | P) | 8 |
| II) | A – B | Q) | 9 |
| III) | $A^B$ | R) | 1 |
| IV) | $B^A$ | S) | 5 |
| | | T) | 3 |

(A) I – S; II – R; III – Q; IV – P (B) I – S; II – T; III – P; IV – Q

(C) I – Q; II – P; III – T; IV – S (D) I – Q; II – P; III – S; IV – R

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q25

**Surds & Indices**

JEE Mains · Maths · Numerical · marking +4 / −1 · marks lost 4

Top 100: **44%** got it right · Campus: **21%** (13 of 63)

**Within reach**

**You: Skipped**

25. If $\left(2+\sqrt{3}\right)^{x^2-7x+1}+\left(2-\sqrt{3}\right)^{x^2-7x-1}=\frac{2}{2-\sqrt{3}}$, then the sum of values of '$x$' satisfying this equation is equal to_____ ?

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q3

**Sequences & Series** – Product of GP terms; natural-number GP

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5

Top 100: **42%** got it right · Campus: **51%** (32 of 63)

Within reach

You: Wrong

3. If $x_1, x_2, x_3, x_4, x_5, \ldots$ is a geometric progression of natural numbers and $x_1x_2x_3x_4 = 64$, then complete set of solutions of $x_4$ is

(A) $\{1\}$ (B) $\{8\}$ (C) $\{1, 8\}$ (D) cannot be determined

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q4

**Sequences & Series** – Identify AP / GP / HP from a surd expression

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 4

Top 100: **42%** got it right · Campus: **19%** (12 of 63)

Within reach

You: Skipped

4. If $a, b, c$ are in A.P., then $\frac{1}{\sqrt{a}+\sqrt{b}}, \frac{1}{\sqrt{a}+\sqrt{c}}, \frac{1}{\sqrt{b}+\sqrt{c}}$ are in

(A) A.P. (B) G.P. (C) H.P. (D) Neither A.P nor G.P nor H.P

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q5

**Quadratic Equations** – Range of a rational quadratic expression

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5

Top 100: **21%** got it right · Campus: **6%** (4 of 63)

Tough one

You: Wrong

5. If $x$ is real, then $\frac{x^2+2x+c}{x^2+4x+3c}$ can take all real values if

(A) $0 < c < 2$ (B) $0 \le c \le 1$

(C) $-1 < c < 1$ (D) $0 < c < 1$

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

**Q16** **Quadratic Equations – Roots in GP, condition on discriminant**

JEE Mains · Maths · Single correct · marking +4 / −1 · marks lost 5

Top 100: **19%** got it right · Campus: **3%** (2 of 63)

**Tough one**

**You: Wrong**

16. Let $\alpha, \beta$ be the roots of the equation $ax^2 + bx + c = 0$ and $\Delta = b^2 - 4ac$. If $\alpha + \beta, \alpha^2 + \beta^2, \alpha^3 + \beta^3$ are in G.P., then

(A) $\Delta = 0$ (B) $\Delta \neq 0$ (C) $b\Delta = 0$ (D) $c\Delta = 0$

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## JEE Mains · Physics – questions to revisit

**15** questions · easy misses first
cropped from the question paper

### Q28

**Vectors – Resultant ⊥ smaller vector; find magnitudes**
JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 4
Top 100: **77%** got it right · Campus: **30%** (19 of 63)

**Easy miss**
**You: Skipped**

28. The simple sum of two co-initial vectors is 16 units . Their vector sum is 8 units. The resultant of the vectors is perpendicular to the smaller vector . The magnitudes of the two vectors are

(A) 22units and 14 units
(B) 4 units and 12 units
(C) 6 units and 10 units
(D) 8 units and 12 units

**Why did I lose this question?** (write the reason and the correct approach)

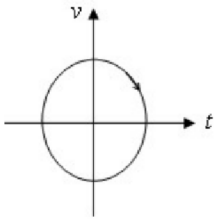
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

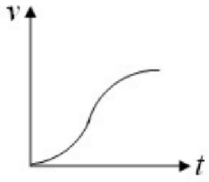
### Q33

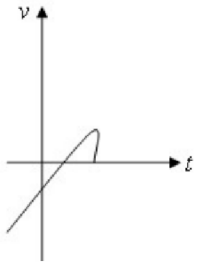
**Kinematics – Which v-t graph can show real one-dimensional motion**
JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5
Top 100: **75%** got it right · Campus: **65%** (41 of 63)

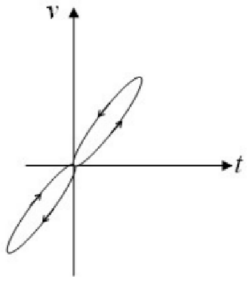
**Easy miss**
**You: Wrong**

33. From following graph which of these possibly represents one-dimensional motion of the particle.

(A) 


(B) 


(C) 


(D) 


**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q29

**Vectors – Angles between three vectors summing to zero**

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **62%** got it right · Campus: **24%** (15 of 63)

**Within reach**

**You: Wrong**

29. Given: $\vec{a}+\vec{b}+\vec{c}=0$ . Out of the three vectors $\vec{a},\vec{b}$ and $\vec{c}$ , two are equal in magnitude. The magnitude of the third vector is $\sqrt{2}$ times that of either of the two having equal magnitude. The angles between the vector are

(A) $90^0,135^0,135^0$ (B) $30^0,60^0,90^0$

(C) $45^0,45^0,90^0$ (D) $45^0,60^0,90^0$

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q40

**Laws of Motion – Velocity over fixed distance vs mass (v ∝ 1/√m)**

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **62%** got it right · Campus: **30%** (19 of 63)

**Within reach**

**You: Wrong**

40. A constant force acts on a mass m at rest. Velocity acquired in travelling a fixed distance is directly proportional to

(A) $\sqrt{m}$ (B) m (C) $1/\sqrt{m}$ (D) $1/m$

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q50

**Laws of Motion**

JEE Mains · Physics · Numerical · marking +4 / −1 · marks lost 4

Top 100: **44%** got it right · Campus: **17%** (11 of 63)

**Within reach**

**You: Skipped**

50. A constant force acts on a body of mass 1kg for 5 seconds, so that the body begins to move with zero initial velocity on a smooth horizontal plane. After 5 seconds the force ceases to act. In the next 5 second, the body describes 100m. What is force ( in Newton) that acted on the body

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q27

**Vectors** – Vector components under axis rotation

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 4

Top 100: **43%** got it right · Campus: **6%** (4 of 63)

**Within reach**

**You: Skipped**

27. The components of a vector along x and y directions are (n+1) and 1 respectively . If the co-ordinate system is rotated by an angle $\theta = 60^0$, then the components change to n and 3. The value of n is

(A) 2 (B) $\cos 60^0$ (C) $\sin 60^0$ (D) 3.5

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q35

**Kinematics** – a = kv² – integrate to get v(s)

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 4

Top 100: **41%** got it right · Campus: **5%** (3 of 63)

**Within reach**

**You: Skipped**

35. If a particle accelerates with $a = kv^2$ and initial velocity = $u$, then find velocity after displacement s

(A) $v = ue^{ks}$ (B) $v = u^{ks}e^{s}$ (C) $v = u^{k}e^{s}$ (D) $v = eu^{ks}$

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q41

**Laws of Motion** – Newton's 3rd law – fan blowing on own sail

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **31%** got it right · Campus: **11%** (7 of 63)

**Tough one**

**You: Wrong**

41. On a stationary sail boat, air is blown at the sails from a fan attached to the boat. The boat will:

(A) Move forward (B) move backward

(C) spin around (D) remain stationary

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q42 — Laws of Motion – Extra tension when bird leaves the wire

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **31%** got it right · Campus: **14%** (9 of 63)

**Tough one**

**You: Wrong**

42. A bird dismounts on a stretched telegraph wire. The additional tension produced in the wire is

(A) Zero

(B) greater than the weight of the bird

(C) Less than the weight of the bird

(D) equal to the weight of the bird

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q38 — Kinematics – Resistance ∝ v² – time inside plank

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 4

Top 100: **27%** got it right · Campus: **11%** (7 of 63)

**Tough one**

**You: Skipped**

38. Having gone through a plank of thickness $h$, a bullet changed its velocity from $v_0$ to $v$. Find the time of motion of the bullet in the plank assuming resistance force to be proportional to the square of the velocity

(A) $\left[\frac{h(v_0-v)}{vv_0 l\,n\frac{v_0}{v}}\right]$ (B) $\left[\frac{(v_0-v)}{hvv_0 l\,n\left(\frac{v_0}{v}\right)}\right]$ (C) $\left[\frac{vv_0 \ln\left(\frac{v_0}{v}\right)}{h(v_0-v)}\right]$ (D) $\left[\frac{\ln\frac{v_0}{v}}{vv_0(v_0-v)}\right]$

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q36 — Kinematics – a(t) with modulus – area under a-t

JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **24%** got it right · Campus: **5%** (3 of 63)

**Tough one**

**You: Wrong**

36. Starting from rest a particle moves in a straight line with acceleration $a=\{2+|t-2|\}\,m/s^2$. Velocity of the particle at the end of 4s will be

(A) 16 m/s (B) 20 m/s (C) 8 m/s (D) 12 m/s

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q43 — Laws of Motion – Normal reaction across an inclined contact

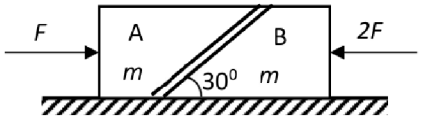
JEE Mains · Physics · Single correct · marking +4 / −1 · marks lost 5

Top 100: **20%** got it right · Campus: **10%** (6 of 63)

**Tough one**

**You: Wrong**

43. Two blocks ‘A” and ‘B’ each of mass ‘m’ are placed on a smooth horizontal surface. Two horizontal forces F and 2F are applied on both the blocks ‘A’ and ‘B’ respectively as shown in figure. The block A does not slide on block B. Then the normal reaction acting between the two blocks is




(A) F (B) $F/2$ (C) $F/\sqrt{3}$ (D) 3F

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q47 — Vectors – Net displacement from multi-leg journey

JEE Mains · Physics · Numerical · marking +4 / −1 · marks lost 4

Top 100: **15%** got it right · Campus: **3%** (2 of 63)

**Tough one**

**You: Skipped**

47. In going from one city to another , a car travels 75 km north, 60 km north –west and 20 km east. The magnitude(in km) of the vector drawn between the two cities is $\left(Take 1/\sqrt{2}=0.7\right)$

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q48

**Kinematics – Distance from a-t graph**

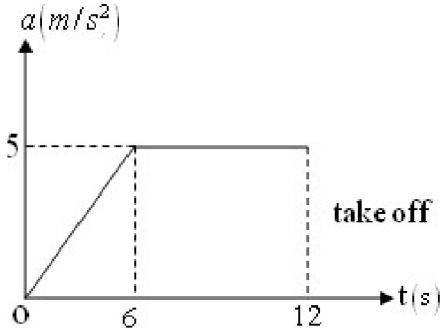
JEE Mains · Physics · Numerical · marking +4 / −1 · marks lost 5

Top 100: **15%** got it right · Campus: **3%** (2 of 63)

**Tough one**

**You: Wrong**

48. An experiment on the take off performance of an aeroplane shows that the acceleration varies as shown in the fig and it takes 12 sec to take off from a rest position.The distance along the runway covered by the aeroplane is(in m).




**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q49

**Kinematics – Free fall then deceleration – find height**

JEE Mains · Physics · Numerical · marking +4 / −1 · marks lost 4

Top 100: **15%** got it right · Campus: **2%** (1 of 63)

**Tough one**

**You: Skipped**

49. A parachutist after bailing out falls 50m without friction. When parachute opens, it decelerates at $2m/\sec^2$. He reaches the ground with a speed of 3m/s. The height he bail out is nearly (in m) (Take g=10m/sec$^2$) .

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## JEE Mains · Chemistry — questions to revisit

**7** questions · easy misses first
cropped from the question paper

### Q54

**Mole Concept & Basics** — **Minimum molecular mass of insulin from 3.4% S**
JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 4
Top 100: **67%** got it right · Campus: **25%** (16 of 63)

**Within reach**
**You: Skipped**

54. Insulin contains 3.4% sulphur. What will be the minimum molecular weight of insulin ?
(A) 94.176 (B) 1884 (C) 941.176 (D) 976

**Why did I lose this question?** (write the reason and the correct approach)
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

### Q55

**Mole Concept & Basics** — **Laws of chemical combination**
JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5
Top 100: **66%** got it right · Campus: **17%** (11 of 63)

**Within reach**
**You: Wrong**

55. 12 g carbon combines with 64 g sulphur to form $CS_2$. 12 g carbon also combines with 32 g oxygen to form $CO_2$. 10 g sulphur combines with 10 g oxygen to form $SO_2$. These data illustrate the -
(A) Law of multiple proportions
(B) Law of definite proportions
(C) Law of reciprocal proportions
(D) Law of gaseous volumes.

**Why did I lose this question?** (write the reason and the correct approach)
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

### Q65

**Periodic Classification** — **Largest radius among isoelectronic ions**
JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 4
Top 100: **64%** got it right · Campus: **33%** (21 of 63)

**Within reach**
**You: Skipped**

65. Which of the following has largest radius ?
(A) $Mg^{2+}$ (B) $Na^{+}$ (C) $O^{2-}$ (D) $F^{-}$

**Why did I lose this question?** (write the reason and the correct approach)
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q75

**Chemical Bonding – Hybrid orbitals + lone pairs in $NH_3$**
JEE Mains · Chemistry · Numerical · marking +4 / −1 · marks lost 4
Top 100: **60%** got it right · Campus: **30%** (19 of 63)

**Within reach**
**You: Skipped**

75. According to VBT, consider the molecule $NH_3$. The central nitrogen atom undergoes hybridisation before bond formation.
If the number of hybrid orbitals formed is x, and the number of lone pairs present on nitrogen is y, find:
x + y

**Why did I lose this question?** (write the reason and the correct approach)
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q58

**Atomic Structure**
JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5
Top 100: **57%** got it right · Campus: **16%** (10 of 63)

**Within reach**
**You: Wrong**

58. The ratio of the radii of first orbits of H, $He^+$ and $Li^{2+}$ is -
(A) 1 : 2 : 3 (B) 6 : 3 : 2 (C) 1 : 4 : 9 (D) 9 : 4 : 1

**Why did I lose this question?** (write the reason and the correct approach)
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q59

**Chemical Bonding – VBT, orbital overlap, hybridisation statements**
JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 4
Top 100: **50%** got it right · Campus: **29%** (18 of 63)

**Within reach**
**You: Skipped**

59. Statement I: According to VBT, a covalent bond is formed by the overlap of half-filled atomic orbitals containing electrons with opposite spins.
Statement II: Greater the extent of effective orbital overlap, stronger is the covalent bond formed.
Statement III: Hybridisation involves the mixing of atomic orbitals of comparable energy to form equivalent hybrid orbitals.
**Which of the following is correct?**
(A) Only Statement I is correct
(B) Statements I and II are correct
(C) Statements I, II and III are correct
(D) Statements II and III are correct

**Why did I lose this question?** (write the reason and the correct approach)
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

**Q67** **Chemical Bonding – Strongest ionic bond (charge and size)**
JEE Mains · Chemistry · Single correct · marking +4 / −1 · marks lost 5
Top 100: **32%** got it right · Campus: **13%** (8 of 63)

**Tough one**
**You: Wrong**

67. Which combination will give strongest ionic bond?

(A) $K^+$ and $Cl^-$ (B) $K^+$ and $O^{2-}$

(C) $Ca^{2+}$ and $Cl^-$ (D) $Ca^{2+}$ and $O^{2-}$

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

# JEE Advanced – question by question

80 / 171 · 46.8%
09 Oct 2026

## 6 Every question at a glance

Right · Wrong · Skipped · Deleted (no marks) · Easy miss · Within reach · Tough one

| | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Maths | 1 | 2 | 3 | **4** | 5 | 6 | 7 | 8 | 9 | **10** | 11 | 12 | **13** | ~~14~~ | **15** | **16** | ~~17~~ |
| Physics | 18 | 19 | **20** | 21 | 22 | **23** | **24** | **25** | 26 | **27** | **28** | **29** | **30** | **31** | **32** | **33** | **34** |
| Chemistry | **35** | **36** | 37 | **38** | **39** | 40 | **41** | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | ~~50~~ | **51** |

## 7 Questions you lost – what to fix first

Every question you got wrong or skipped, one table per subject, easiest marks first. **Top 100** = the batch's top 100 across all campuses; **Campus** = all 63 students of this program on campus.

**Easy miss** 70%+ of the top 100 got it right – easiest marks to win back · **Within reach** 40–69% · **Tough one** below 40%

### Maths

**5** of 15 questions lost · **18** marks · ● 3 within reach · ● 2 tough one

| Q | PRIORITY | CHAPTER – WHAT IT TESTS | TYPE | YOU | MARKS LOST | TOP 100 RIGHT | CAMPUS RIGHT |
|---|---|---|---|---|---|---|---|
| Q4 | Within reach | **Logarithms** – Logs in AP – solve for x | Single correct · +3 / −1 | Wrong | 4 | 68% | 35% |
| Q10 | Within reach | **Quadratic Equations** – Roots of two linked quadratics | Integer · +4 / 0 | Skipped | 4 | 50% | 25% |
| Q15 | Within reach | **Sequences & Series** – AP: ratio of sums of 14 terms | Paragraph · +3 / 0 | Wrong | 3 | 42% | 8% |
| Q16 | Tough one | **Logarithms** – Log equations – A + B + C | Paragraph · +3 / 0 | Wrong | 3 | 18% | 3% |
| Q13 | Tough one | **Quadratic Equations** – Spot the polynomial identity, then sum P(1) … P(2023) | Integer · +4 / 0 | Skipped | 4 | – no data* | 2% |

### Physics

**12** of 17 questions lost · **47** marks (+1 on partly-correct answers) · ● 2 easy miss · ● 3 within reach · ● 7 tough one

| Q | PRIORITY | CHAPTER – WHAT IT TESTS | TYPE | YOU | MARKS LOST | TOP 100 RIGHT | CAMPUS RIGHT |
|---|---|---|---|---|---|---|---|
| Q25 | Easy miss | **Vectors** | Integer · +4 / 0 | Skipped | 4 | 84% | 37% |
| Q24 | Easy miss | **Laws of Motion** – Tensions in two inclined strings | Multiple correct · +4 / −2 | Wrong | 6 | 70% | 44% |
| Q20 | Within reach | **Kinematics** | Single correct · +3 / −1 | Skipped | 3 | 52% | 22% |
| Q23 | Within reach | **Laws of Motion** – Block on block, no friction – motion of m | Multiple correct · +4 / −2 | Wrong | 6 | 51% | 22% |
| Q30 | Within reach | **Kinematics** – a from v(t) at a given position | Integer · +4 / 0 | Wrong | 4 | 43% | 8% |
| Q28 | Tough one | **Laws of Motion** – Spring-connected blocks – acceleration | Integer · +4 / 0 | Wrong | 4 | 25% | 13% |
| Q33 | Tough one | **Kinematics** – Train passage – retardation to avoid collision | Paragraph · +3 / 0 | Skipped | 3 | 15% | 8% |
| Q34 | Tough one | **Kinematics** – Train passage – time of retardation | Paragraph · +3 / 0 | Skipped | 3 | 15% | 10% |
| Q27 | Tough one | **Laws of Motion** – Normal reaction between pushed blocks | Integer · +4 / 0 | Wrong | 4 | 11% | 2% |
| Q31 | Tough one | **Vectors** – Carrom passage – displacement centre → edge | Paragraph · +3 / 0 | Wrong | 3 | 7% | 2% |
| Q29 | Tough one | **Kinematics** | Integer · +4 / 0 | Wrong | 4 | – no data* | 6% |
| Q32 | Tough one | **Vectors** – Carrom passage – displacement edge → hole | Paragraph · +3 / 0 | Wrong | 3 | 6% | 2% |

### Chemistry

**6** of 16 questions lost · **25** marks · ● 3 easy miss · ● 2 within reach · ● 1 tough one

| Q | PRIORITY | CHAPTER – WHAT IT TESTS | TYPE | YOU | MARKS LOST | TOP 100 RIGHT | CAMPUS RIGHT |
|---|---|---|---|---|---|---|---|
| **Q38** | Easy miss | **Chemical Bonding** – Pair of species with the same shape (VSEPR) | Single correct · +3 / −1 | **Wrong** | **4** | **92%** | 49% |
| **Q41** | Easy miss | **Periodic Classification** | Multiple correct · +4 / −2 | **Skipped** | **4** | **92%** | 70% |
| **Q36** | Easy miss | **Atomic Structure** | Single correct · +3 / −1 | **Wrong** | **4** | **72%** | 59% |
| **Q39** | Within reach | **Mole Concept & Basics** – Meaning of 1 gram-atom of nitrogen | Multiple correct · +4 / −2 | **Wrong** | **6** | **66%** | 22% |
| **Q35** | Within reach | **Mole Concept & Basics** – Limiting reagent over two-step reaction | Single correct · +3 / −1 | **Wrong** | **4** | **51%** | 24% |
| **Q51** | Tough one | **Atomic Structure** – Hydrogen-like ion – find Z from photon energies | Paragraph · +3 / 0 | **Skipped** | **3** | **0%** | 5% |

## 8 Chapter scorecard

Your marks on each chapter's questions against the campus average for the same questions. Weakest chapters first within each subject. Chapter marks are approximate (≈): partly-correct answers cannot be traced to a question.

| CHAPTER | QS | YOUR MARKS | YOU VS CAMPUS AVERAGE | CAMPUS AVG | QUESTIONS YOU LOST |
|---|---|---|---|---|---|
| Maths | | | | | |
| **Logarithms** | 4 | ≈ **6**/13 | | 50% | **Q4**, **Q16** |
| **Quadratic Equations** | 7 | ≈ **18**/26 | | 43% | **Q10**, **Q13** |
| **Sequences & Series** | 4 | ≈ **12**/15 | | 56% | **Q15** |
| Physics | | | | | |
| **Laws of Motion** | 4 | ≈ **-4**/16 | | 6% | **Q23**, **Q24**, **Q27**, **Q28** |
| **Kinematics** | 8 | ≈ **10**/27 | | 24% | **Q20**, **Q29**, **Q30**, **Q33**, **Q34** |
| **Vectors** | 5 | ≈ **7**/17 | | 23% | **Q25**, **Q31**, **Q32** |
| Chemistry | | | | | |
| **Chemical Bonding** | 1 | ≈ **-1**/3 | | 41% | **Q38** |
| **Periodic Classification** | 2 | ≈ **3**/7 | | 68% | **Q41** |
| **Atomic Structure** | 4 | ≈ **7**/14 | | 33% | **Q36**, **Q51** |
| **Mole Concept & Basics** | 9 | ≈ **23**/33 | | 44% | **Q35**, **Q39** |

Bar = your % of the chapter's marks | campus average

## 9 JEE Advanced · Maths — questions to revisit

5 questions · easy misses first
cropped from the question paper

### Q4 — Logarithms — Logs in AP — solve for x

JEE Advanced · Maths · Single correct · marking +3 / −1 · marks lost 4
Top 100: **68%** got it right · Campus: **35%** (22 of 63)

**Within reach**
**You: Wrong**

4. If $\log_3 2, \log_3\left(2^x-5\right)$ & $\log_3\left(2^x-\frac{7}{2}\right)$ are in AP then the value of $x$ is?

A)2 B)3 C)4 D)2&3

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

### Q10 — Quadratic Equations — Roots of two linked quadratics

JEE Advanced · Maths · Integer · marking +4 / 0 · marks lost 4
Top 100: **50%** got it right · Campus: **25%** (16 of 63)

**Within reach**
**You: Skipped**

10. Let $a$ and $b$ be roots of the equation $x^2-10cx-11d=0$ and the roots of $x^2-10ax-11b=0$ are $c$ and $d$; then the value of $\frac{a+b+c+d}{605}$ is____?( Given $a\neq b\neq c\neq d\neq 0$ )

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

### Q15 — Sequences & Series — AP: ratio of sums of 14 terms

JEE Advanced · Maths · Paragraph · marking +3 / 0 · marks lost 3
Top 100: **42%** got it right · Campus: **8%** (5 of 63)

**Within reach**
**You: Wrong**

Sum of n terms of two different Arithmetic progressions are in the ratio 5n+4:9n+6 then
15. The ratio of the sum of their first 14 terms is Q, then [100Q] is _____(where [.] is G.I.F)

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q16

**Logarithms – Log equations – A + B + C**

JEE Advanced · Maths · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **18%** got it right · Campus: **3%** (2 of 63)

**Tough one**

**You: Wrong**

### PARAGRAPH-II

Let **A** denote the product $xyz$ where $x, y, z$ are solutions of the system of equations

$\log_3 x = \log_3 5 - \log_3 7, \log_5 y = \log_5 7 - \log_5 3, \log_7 z = \log_7 3 - \log_7 5$

Let **B** denote the sum of square of the solutions of the equation

$\log_2\left(\log_2 x^6 - 3\right) - \log_2\left(\log_2 x^4 - 5\right) = \log_2 3$

Let **C** denote the characteristic of the number K, where

$K = \log_2\left(\log_2 3\right) - \log_2\left(\log_4 3\right) + \log_2\left(\log_4 5\right) - \log_2\left(\log_6 5\right) + \log_2\left(\log_6 7\right) - \log_2\left(\log_8 7\right)$

16. The value of A+B+C is____?

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q13

**Quadratic Equations – Spot the polynomial identity, then sum P(1) ... P(2023)**

JEE Advanced · Maths · Integer · marking +4 / 0 · marks lost 4

Top 100: **no data** got it right · Campus: **2%** (1 of 63)

**Tough one**

**You: Skipped**

13. Let $P(x) = \left(1+\sqrt{2}\right)\left(1+\sqrt{3}\right)\left(x-\sqrt{2}\right)\left(x-\sqrt{3}\right) + \left(1+\sqrt{3}\right)\left(\sqrt{3}+\sqrt{2}\right)(x-1)\left(x-\sqrt{2}\right)$ $-2\left(1+\sqrt{2}\right)\left(\sqrt{3}+\sqrt{2}\right)(x-1)\left(x-\sqrt{3}\right)\ \forall x \in R$. Then the value of $P(1)+P(2)+P(3)+\ldots P(2023)$ is_____?

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## JEE Advanced · Physics – questions to revisit

**12** questions · easy misses first
cropped from the question paper

### Q25 Vectors

JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4
Top 100: **84%** got it right · Campus: **37%** (23 of 63)

**Easy miss**
**You: Skipped**

25. The vectors $\vec{A} = 5\hat{i} - 7\hat{j} - 2\hat{k}$ and $\vec{B} = 2\hat{i} + 2\hat{j} - c\hat{j}$ are perpendicular. Find the value of C.

**Why did I lose this question?** (write the reason and the correct approach)

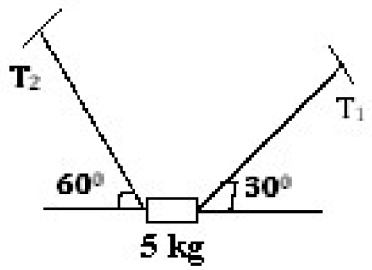
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

### Q24 Laws of Motion – Tensions in two inclined strings

JEE Advanced · Physics · Multiple correct · marking +4 / −2 · marks lost 6
Top 100: **70%** got it right · Campus: **44%** (28 of 63)

**Easy miss**
**You: Wrong**

24. A body of mass 5 kg is suspended by the strings making angles $60^0$ and $30^0$ with the horizontal as shown in the figure($g = 10ms^{-2}$). Then




A) $T_1 = 25N$ B) $T_2 = 25N$ C) $T_1 = 25\sqrt{3}N$ D) $T_2 = 25\sqrt{3}N$

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q20 Kinematics

JEE Advanced · Physics · Single correct · marking +3 / −1 · marks lost 3
Top 100: **52%** got it right · Campus: **22%** (14 of 63)

**Within reach**
**You: Skipped**

20. A particle is projected vertically upwards from the ground at time $t = 0$ and reaches a height $h$ at $t = T$. Greatest height of the particle is

A) $\left(gT^2 + h\right)^2 / 8gT^2$

B) $2\left(gT^2 + h\right)^2 / 8gT^2$

C) $\left(gT^2 + 2h\right)^2 / 8gT^2$

D) $2h$

**Why did I lose this question?** (write the reason and the correct approach)

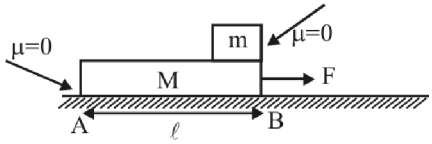
☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q23 Laws of Motion – Block on block, no friction – motion of m

JEE Advanced · Physics · Multiple correct · marking +4 / −2 · marks lost 6
Top 100: **51%** got it right · Campus: **22%** (14 of 63)

**Within reach**
**You: Wrong**

23. In the figure small block is kept on M, then




A) the acceleration of m w.r.t. ground is $\frac{F}{m}$

B) the acceleration of m w.r.t. ground is zero

C) the time taken by m to separate from M is $\sqrt{\frac{2\ell m}{F}}$

D) the time taken by m to separate from M is $\sqrt{\frac{2\ell M}{F}}$

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q30

**Kinematics – a from v(t) at a given position**

JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4

Top 100: **43%** got it right · Campus: **8%** (5 of 63)

**Within reach**

**You: Wrong**

30. A body starts from origin and moves along x axis such that at any instant velocity is $v_t=4t^3-2t$ where t is in second and $v_t$ is in $ms^{-1}$. The acceleration of the particle when it is 2m from the origin is $(20+K)ms^{-2}$, where $K$ is

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q28

**Laws of Motion – Spring-connected blocks – acceleration**

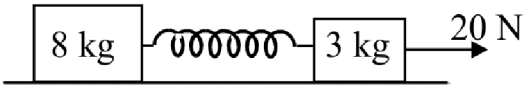
JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4

Top 100: **25%** got it right · Campus: **13%** (8 of 63)

**Tough one**

**You: Wrong**

28. If the acceleration of 3 kg block is 4 $m/s^2$ to the left. What is the magnitude of acceleration of 8 kg block.




**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q33

**Kinematics – Train passage – retardation to avoid collision**

JEE Advanced · Physics · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **15%** got it right · Campus: **8%** (5 of 63)

**Tough one**

**You: Skipped**

**PASSAGE-II:**

The driver of a train A moving with a uniform speed of 144 km/h sights another train B, 1km ahead of him. The train B is moving with a uniform speed of 108 km/h. the driver of the train A immediately applies brakes producing a constant retardation and just manages to avoid a collision.

33. What is the retardation of the train A(in $ms^{-2}$)_____

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q34

**Kinematics – Train passage – time of retardation**

JEE Advanced · Physics · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **15%** got it right · Campus: **10%** (6 of 63)

**Tough one**

**You: Skipped**

**PASSAGE-II:**

The driver of a train A moving with a uniform speed of 144 km/h sights another train B, 1km ahead of him. The train B is moving with a uniform speed of 108 km/h. the driver of the train A immediately applies brakes producing a constant retardation and just manages to avoid a collision.

34. For how long is this retardation produced(in sec)_____

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q27

**Laws of Motion – Normal reaction between pushed blocks**

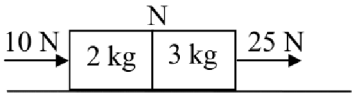
JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4

Top 100: **11%** got it right · Campus: **2%** (1 of 63)

**Tough one**

**You: Wrong**

27. The normal reaction between 2 kg and 3 kg block is




**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q31

**Vectors – Carrom passage – displacement centre → edge**

JEE Advanced · Physics · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **7%** got it right · Campus: **2%** (1 of 63)

**Tough one**

**You: Wrong**

**PASSAGE-1:**

A Corrom board $(4\,ft \times 4\,ft\ square)$ has queen at the centre. The queen , hit by the striker moves to the front edge, rebounds and goes in the hole behind the striking line.

Answer the following question.

31. The magnitude of the displacement of the queen from the centre to the front edge is $\frac{2}{3}\sqrt{a}\,ft$ , the value of a is _____

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q29

**Kinematics**

JEE Advanced · Physics · Integer · marking +4 / 0 · marks lost 4

Top 100: **no data** got it right · Campus: **6%** (4 of 63)

**Tough one**

**You: Wrong**

29. A lead ball is dropped freely into a lake from a diving board 5 m above the water. It strikes the water with a certain velocity and then reaches the bottom with the same constant velocity in 5 second after it is dropped.
Take g=10 $ms^{-2}$ calculate the average velocity of the ball in $ms^{-1}$ just before reaches to the bottom from the freely dropped point .

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q32

**Vectors – Carrom passage – displacement edge → hole**

JEE Advanced · Physics · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **6%** got it right · Campus: **2%** (1 of 63)

**Tough one**

**You: Wrong**

**PASSAGE-1:**

A Corrom board $(4\,ft \times 4\,ft\ square)$ has queen at the centre. The queen , hit by the striker moves to the front edge, rebounds and goes in the hole behind the striking line.
Answer the following question.

32. The magnitude of the displacement of the queen from the front edge to the hole is $\frac{4}{3}\sqrt{b}\ ft$, the value of b is_____

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## JEE Advanced · Chemistry — questions to revisit

6 questions · easy misses first
cropped from the question paper

### Q38 — Chemical Bonding — Pair of species with the same shape (VSEPR)

JEE Advanced · Chemistry · Single correct · marking +3 / −1 · marks lost 4
Top 100: **92%** got it right · Campus: **49%** (31 of 63)

**Easy miss**
**You: Wrong**

38. The pair of species with similar shape is

(A) $PCl_3$, $NH_3$ (B) $CF_4$, $SF_4$ (C) $PbCl_2$, $CO_2$ (D) $PF_5$, $IF_5$

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

### Q41 — Periodic Classification

JEE Advanced · Chemistry · Multiple correct · marking +4 / −2 · marks lost 4
Top 100: **92%** got it right · Campus: **70%** (44 of 63)

**Easy miss**
**You: Skipped**

41. Which of the following trends of atomic sizes are correct ?

(A) F > N > O > C (B) Rb > K > Na > Li
(C) Be > B > C > N (D) Ne >He > Ar > Kr

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

### Q36 — Atomic Structure

JEE Advanced · Chemistry · Single correct · marking +3 / −1 · marks lost 4
Top 100: **72%** got it right · Campus: **59%** (37 of 63)

**Easy miss**
**You: Wrong**

36. As we go away from the nucleus, the energy difference between the successive energy levels -

(A) increases (B) decreases
(C) remains constant (D) first increases, then decreases

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q39

**Mole Concept & Basics – Meaning of 1 gram-atom of nitrogen**

JEE Advanced · Chemistry · Multiple correct · marking +4 / −2 · marks lost 6

Top 100: **66%** got it right · Campus: **22%** (14 of 63)

**Within reach**

**You: Wrong**

39. 1 g atom of nitrogen represents:

(A) $6.02 \times 10^{23}$ $N_2$ molecules

(B) 22.4 litre of $N_2$ at N.T.P.

(C) 11.2 litre of $N_2$ at N.T.P.

(D) 14 g of nitrogen

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q35

**Mole Concept & Basics – Limiting reagent over two-step reaction**

JEE Advanced · Chemistry · Single correct · marking +3 / −1 · marks lost 4

Top 100: **51%** got it right · Campus: **24%** (15 of 63)

**Within reach**

**You: Wrong**

35. 5 moles of A, 6 moles of Z and mixed with sufficient amount of C to produce final product F. How many maximum moles of 'F' can be produced as per the given sequence of reaction ?

$A + 2Z \rightarrow B$ ; $B + C \rightarrow Z + F$

(A) 3 (B) 2 (C) 4 (D) 5

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## Q51

**Atomic Structure – Hydrogen-like ion – find Z from photon energies**

JEE Advanced · Chemistry · Paragraph · marking +3 / 0 · marks lost 3

Top 100: **0%** got it right · Campus: **5%** (3 of 63)

**Tough one**

**You: Skipped**

**PARAGRAPH-II**

When electron jumps from lower orbit to higher orbit energy is gained and when electron jumps from higher orbit to lower orbit energy is released.

51. A hydrogen like species (atomic number Z) is present in a higher excited state of quantum number n. This excited atom can make a transition to the first excited state by successive emission of two photons of energies 10.20 eV and 17.0 eV respectively. Alternatively the atom from the same excited state can make a transition to the second excited state by successive emission of two photons of energy 4.25 eV and 5.95 eV respectively. Determine the value of Z.

**Why did I lose this question?** (write the reason and the correct approach)

☐ Did not know the concept ☐ Calculation mistake ☐ Misread the question ☐ Ran out of time ☐ Guessed

## 10 How to read this report

**Where the numbers come from.** Marks, percentages, All-India / State / Campus ranks, subject marks, subject All-India ranks and the wrong / skipped questions are the official results in the institute's *All India Marks Analysis* (graded with the final answer key). Maximum marks leave out deleted questions — JEE Mains: Q20 — JEE Advanced: Q14, Q17, Q50. **Program rank** = rank in Pragya G8 (EXI) Achievers across all campuses, from the batch's question-wise sheet († JEE Advanced: that sheet was graded before the final key, so this rank may differ slightly). **Top 100** = the batch's top 100 across all campuses, from the same question-wise sheet; * where it has no data for a question, the campus figure decides the priority. In JEE Advanced that sheet predates the final key, so its figures for the revised questions Q25, Q48, Q51 use the old answers. **Campus** = all students of this program on campus. **Priority:** Easy miss = 70%+ of the top 100 got it right; Within reach = 40–69%; Tough one = below 40%. **Marks lost** on a question = its marks plus the negative mark if answered wrong, its marks if skipped. In JEE Advanced, 1 mark went on multiple-correct answers that were only partly correct. **Weekly-test average** = your average over the weekly Mains tests (WTM) or Advanced tests (WTA) before this Part Test, from the campus averages sheet. The weekly Advanced tests are marked out of 240; their averages are scaled to 180 to match this paper (× 180/240, so 80 per subject becomes 60). Percentages are unchanged.